International Baccalaureate®
Baccalauréat International
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BIOLOGY
HIGHER LEVEL
PAPER 1

SPECIMEN PAPER

1 hour

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- The maximum mark for this examination paper is *[40 marks]*.

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1. What structures are found in eukaryotes but not in prokaryotes?

A. Naked DNA and mitochondrion

B. Golgi apparatus and 70S ribosomes

C. 70S ribosomes and nuclear membrane

D. Mitochondrion and nuclear membrane

2. Membrane proteins of mice cells were marked with green and membrane proteins of human cells were marked with red. The cells were fused together. What would be seen after two hours?

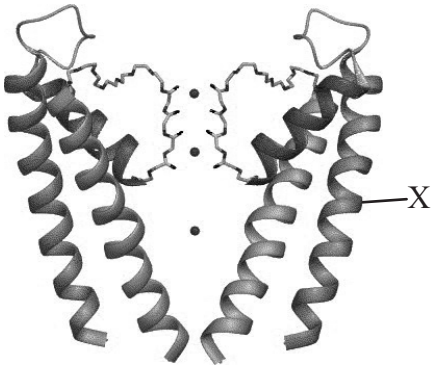
A. Red and green markers fully mixed

B. All red markers and no green markers

C. Half of the new cell with green markers and the other half with red

D. Red markers on the inside of the membrane and green markers on the outside

Question 3 and question 4 refer to the following diagram. It shows a potassium channel in an axon membrane. The three dots in the channel represent potassium ions.



[Source: adapted from SY Noskov and B Roux, (2006), *Biophysical Chemistry*, **124** (3), pages 279–291]

3. What is the function of this potassium channel?

A. Facilitated diffusion of potassium ions into the axon

B. Passage of potassium ions by active transport out of the axon

C. Facilitated diffusion of potassium ions out of the axon

D. Passage of potassium ions by active transport into the axon

4. What structure is labelled X?

A. DNA

B. Beta pleated sheet

C. Alpha helix

D. Prosthetic group

5. In the experiments performed by Meselson and Stahl, *E. coli* were grown for many generations in ^{15}N then for one generation in ^{14}N. What results for the DNA of the last generation showed that replication was semi-conservative?

A. Both strands containing only ^{15}N

B. Both strands containing only ^{14}N

C. One strand containing only ^{15}N and one strand containing only ^{14}N

D. Both strands containing a mixture of ^{15}N and ^{14}N in equal amounts

6. In 1828, Friedrich Wöhler artificially produced urea (organic compound) by an internal rearrangement of the atoms of ammonium cyanate (inorganic compound). What important principle did this experiment show?

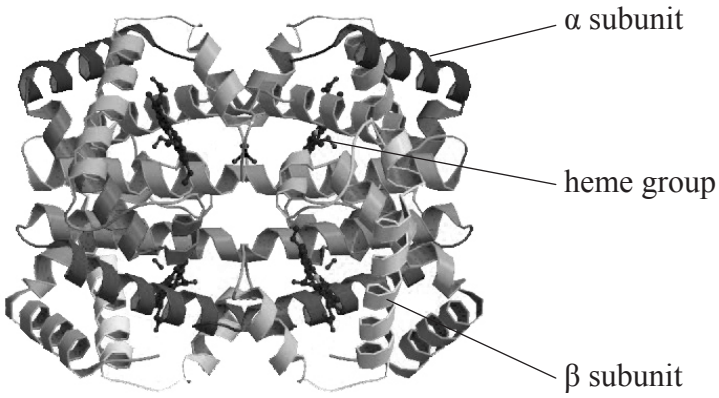
A. Organic compounds can be synthesized without the need of a vital force.

B. Organic compounds can only be formed from pre-existing inorganic compounds.

C. Inorganic compounds are essential for living organisms.

D. Organic compounds are more simple substances than inorganic compounds.

7. The diagram shows the molecular structure of human hemoglobin as found in Protein Data Bank.



[Source: www.rcsb.org/pdb]

What characteristic of hemoglobin allows it to be considered a protein with quaternary structure?

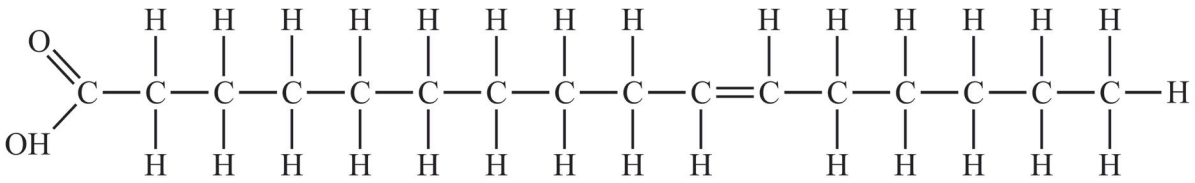
A. It contains many beta sheets.

B. It consists of polypeptide subunits and heme groups.

C. It allows bonding to oxygen atoms.

D. It contains histidine (His) residues.

8. What type of fatty acid is shown in this diagram?



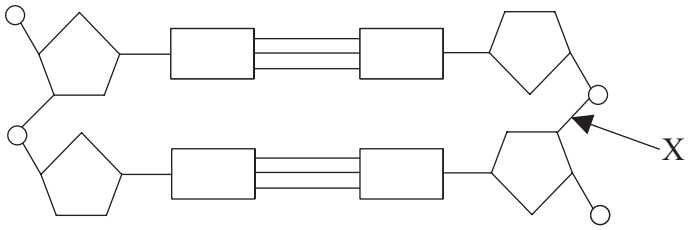
A. Saturated cis

B. Saturated trans

C. Unsaturated cis

D. Unsaturated trans

9. The diagram shows part of a DNA molecule.



What type of bond is labelled X?

A. Covalent bond

B. Hydrogen bond

C. Peptide bond

D. Semi-conservative bond

10. The following alignment represents part of the sequence of a gene in two species, the mouse (*Mus musculus*) and woolly monkey (*Lagothrix lagotricha*).

Mouse MGDVEKGKKIFVMKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANKNK

Woolly monkey MGDVEKGKRIFIMKCSQCHTVEKGGKHKTGXNLHGLFGRKTGQASGYTYTEANKNK

What term is used for different forms of a gene such as these?

A. Loci

B. Alleles

C. Homologues

D. Heterologues

11. Rice has 24 chromosomes in diploid cells while humans have 46. What is a valid conclusion from these data?

A. Plants always have fewer chromosomes than animals.

B. More chromosomes means the species evolved more.

C. The number of chromosomes is a unique characteristic of species.

D. The number of chromosomes present in an organism is random.

12. The electron micrograph shows the chromosomes in one stage of meiosis in rice. Image I shows the whole cell with a boxed region. Image II shows a magnified view of the boxed region.

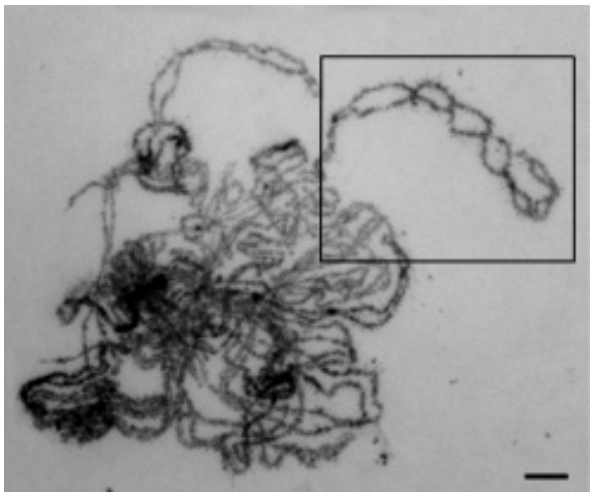


Image I

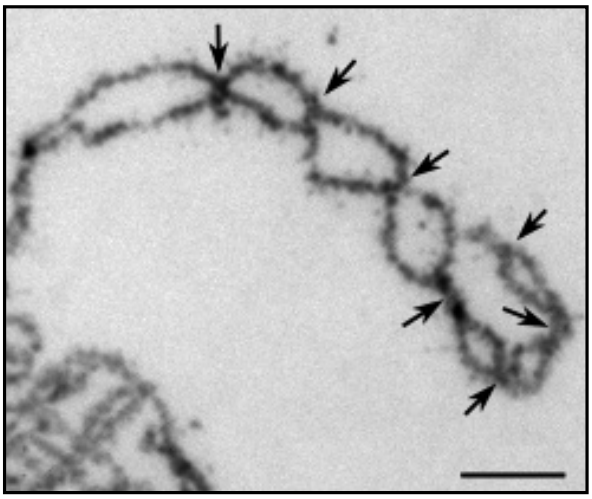


Image II

[Source: adapted from M Wang, *et al*., (2010), Plant Cell, **22** (2), pages 417–430]

What do the arrows indicate?

A. Condensation

B. Chiasmata

C. Centrioles

D. Centromeres

13. Huntington's disease is a neurological disorder caused by the repetition of the amino acid glutamine in the protein Huntingtin. The higher the number of repetitions of glutamine, the earlier the onset of the disease. What type of disease is it?

A. It is an inherited disease.

B. It is a nutritional disease.

C. It is a sexually transmitted disease.

D. It is a sex-linked disease.

14. The table shows the genetic code.

	U	**C**	**A**	**G**	
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	STOP	STOP	**A**
	Leu	Ser	STOP	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**

Which triplet coding for methionine represents the start codon?

A. AUG

B. UAC

C. UGA

D. ACU

15. What is a mesocosm?

A. A lake where experiments are performed in uncontrolled conditions

B. A small area where parts of the natural environment are kept under controlled conditions

C. An experimental area in a laboratory

D. An ocean

16. Which group of organisms converts carbon into a form that is available to primary consumers?

A. Decomposers

B. Saprotrophs

C. Detritivores

D. Producers

17. What is a consequence of a global temperature rise on arctic ecosystems?

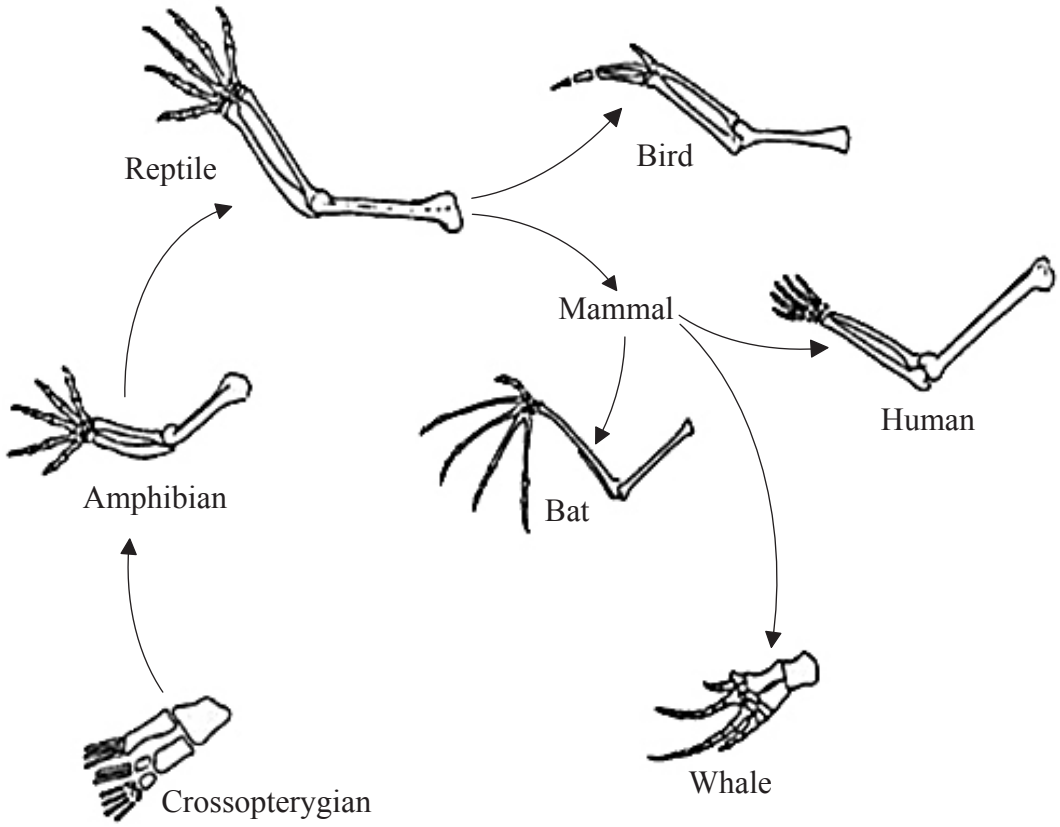
A. Decrease in CO_2 released from decomposing detritus

B. Increase in the greenhouse effect

C. Decrease in ocean level

D. Increase in pest species

18. The limbs of many vertebrates have a similar structure (the pentadactyl limb). In the diagram, all of the organisms have humerus, radius and ulna bones.



[Source: adapted from http://hscbiology.wordpress.com]

What hypothesis is currently accepted to account for the similarity in the limbs?

A. Organisms pass on characteristics that they acquire during their lifetime.

B. The pentadactyl limb is an ideal design for a broad range of purposes.

C. All of the organisms have descended from a common ancestor.

D. Convergent evolution has resulted in each organism finding a similar solution to a mechanical problem.

19. What are the three domains of living organisms?

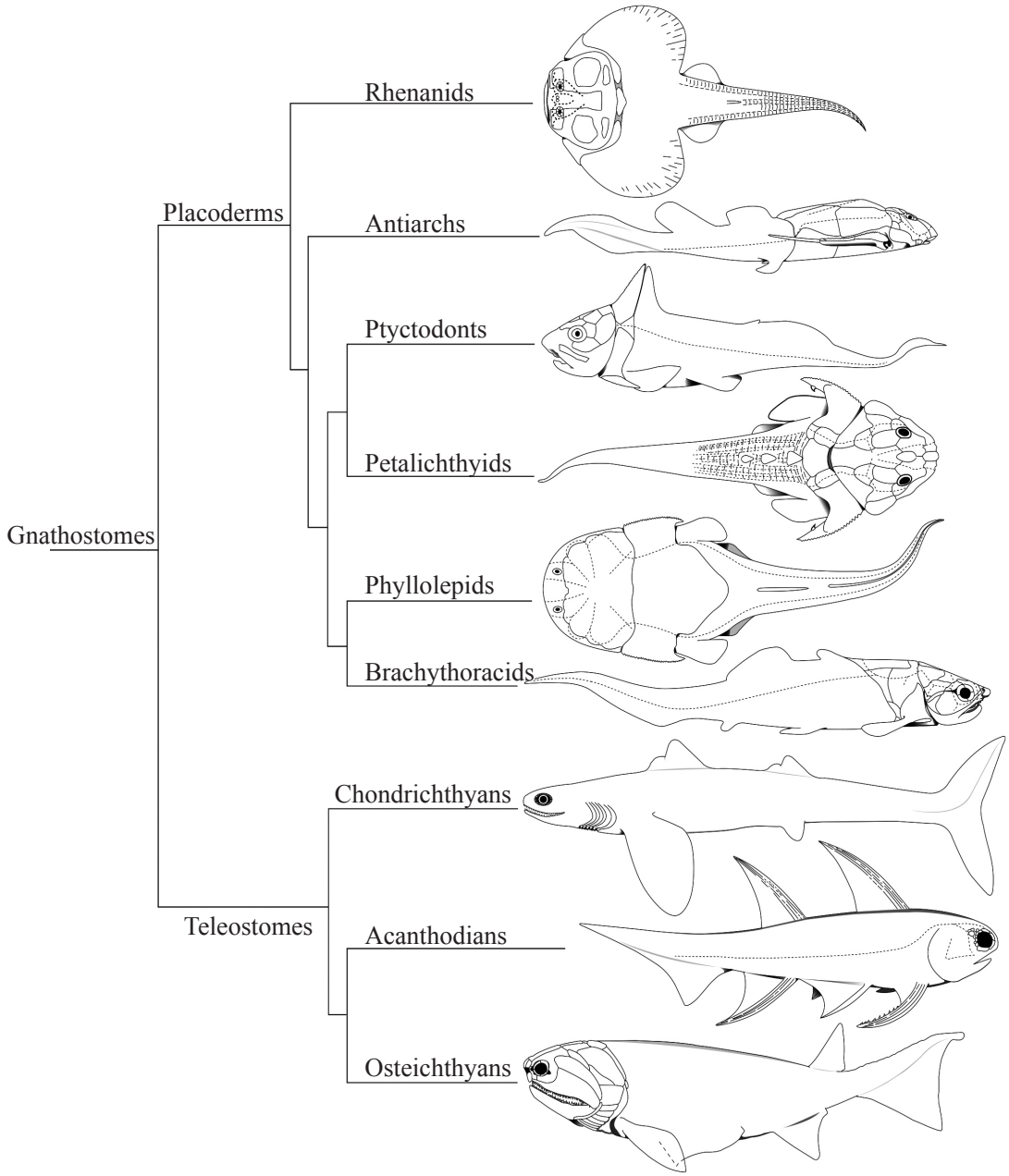
A. Classes, orders and families

B. Bacteria, eukaryotes and viruses

C. Archaea, eubacteria and eukaryotes

D. Decomposers, producers and consumers

20. The cladogram shows the phylogenic relationships in jawed vertebrates.



[Source: adapted from R K Carr and G L Jackson, (2008), *Guide to the Geology and Paleontology of the Cleveland Member of the Ohio Shale*, Ohio Geological Survey Guidebook 22, Chapter 5]

What can be deduced about Brachythoracids from this cladogram?

A. They evolved from Placoderms.

B. They gave rise to Gnathostomes.

C. They evolved at the same time as Ptyctodonts.

D. They differ from Phylollepids by only one mutation.

21. What is a similarity between arteries and capillaries?

A. They both have elastic tissue.

B. They both have smooth muscle cells.

C. Neither has collagen fibres in their walls.

D. Neither has valves.

22. What effect does HIV have on the immune system?

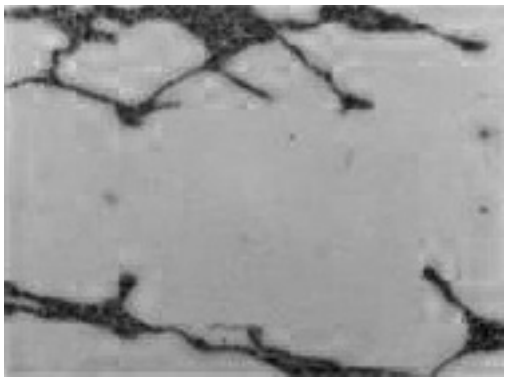
A. It prevents leucocytes from fighting bacteria by phagocytosis.

B. It causes excessive production of leucocytes in bone marrow.

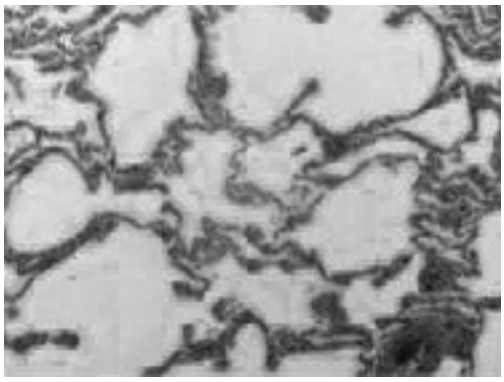
C. It destroys antibodies produced by leucocytes.

D. It reduces antibody production by lowering the number of leucocytes.

23. Emphysema is a long-term, progressive disease that causes shortness of breath. The electron micrographs show the alveoli of a patient with emphysema and the alveoli of a normal person.



Emphysema (magnification × 200)



Normal lung (magnification × 200)

[Source: with permission of *Nature Communications*, 21 May (2013)]

What can be observed in the alveoli of a person suffering from emphysema?

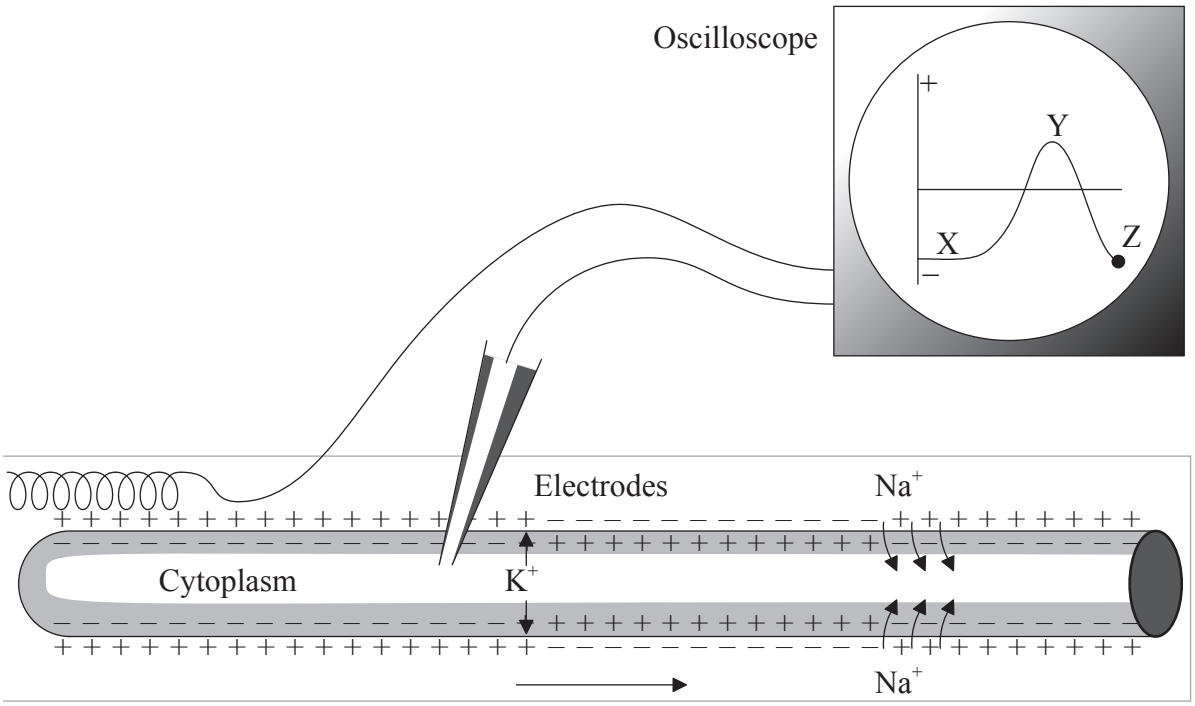
A. Large surface area, large air spaces and many capillaries

B. Small surface area, large air spaces and few capillaries

C. Small surface area, small air spaces and few capillaries

D. Large surface area, small air spaces and many capillaries

24. The diagram shows the results obtained with an oscilloscope attached to a neuron.



[Source: adapted from www.topbiomedical.com]

Why is the change in the oscilloscope occurring between X and Y?

A. Hyperpolarization

B. Hypopolarization

C. Repolarization

D. Depolarization

25. Which response takes place when blood glucose levels are low?

A. Glucagon is released from the α cells of the pancreatic islets.

B. Glucagon is released from the β cells of the pancreatic islets.

C. Insulin is released from the α cells of the pancreatic islets.

D. Insulin is released from the β cells of the pancreatic islets.

26. In the Hershey and Chase experiment what observation led to the conclusion that DNA was the genetic material?

A. Phages containing radioactive phosphorus were unable to infect the bacteria.

B. Phages containing radioactive sulfur were unable to infect the bacteria.

C. Radioactive phosphorus was found in the pellet.

D. Radioactive sulfur was found in the pellet.

27. Some regions of DNA do not code for proteins. What is coded for by some of these regions?

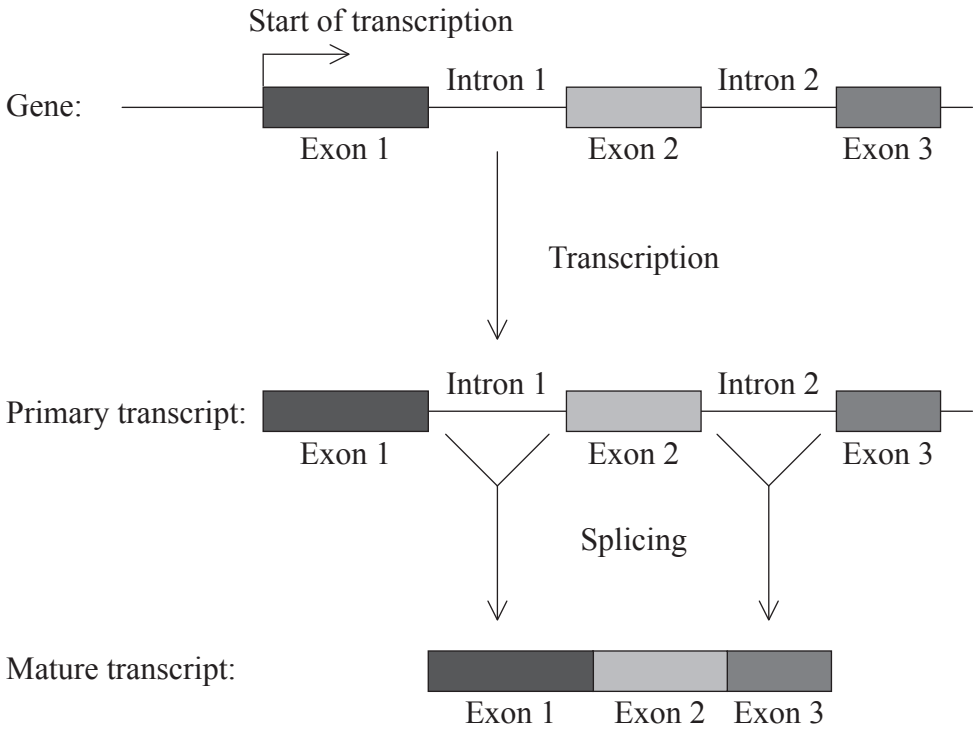
A. tRNA

B. mRNA

C. cDNA

D. Enzymes

28. The diagram shows the post-transcription changes occurring in RNA.



[Source: adapted from http://faculty.uca.edu]

Where in the cell does the splicing process occur in eukaryotes?

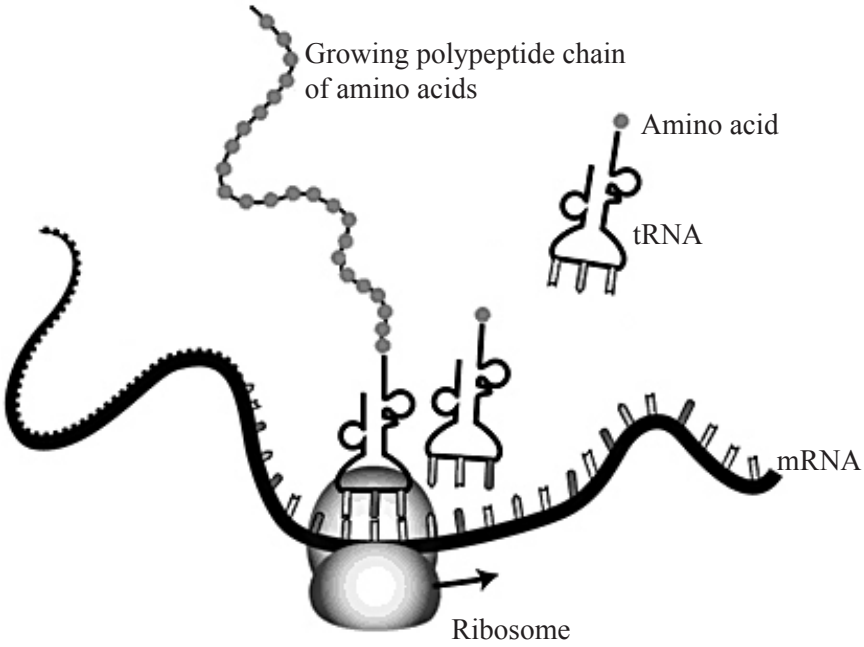
A. Nucleus

B. Ribosomes

C. Cytoplasm

D. Golgi apparatus

29. The diagram shows the translation of an mRNA molecule.



[Courtesy: National Human Genome Research Institute]

A tRNA molecule with anticodon CAG carries the amino acid valine. Which codon of mRNA will the tRNA join?

A. CTG

B. CAG

C. GTC

D. GUC

30. How does a competitive inhibitor interact with an enzyme?

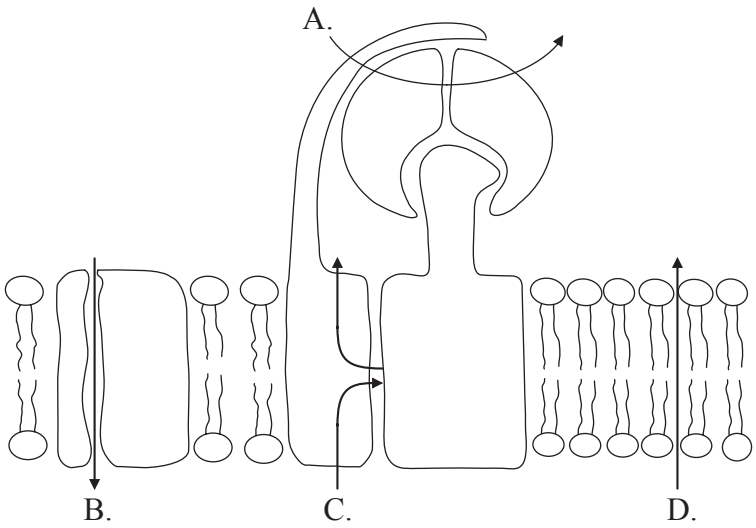
A. It binds to the active site, denaturing the enzyme.

B. It binds to the active site, preventing substrate binding.

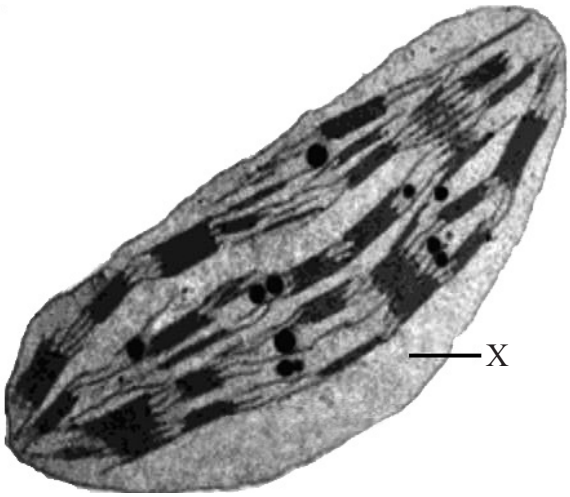
C. It binds to the allosteric site, causing competition with the substrate.

D. It binds to an allosteric site, causing conformational change of the enzyme.

31. The diagram represents components of the cristae in mitochondria. Which arrow indicates how protons (H^+) move to generate ATP directly?



32. The image shows an electron micrograph of a chloroplast.



[Source: www.uic.edu]

What is a function of X?

A. Carbon fixation

B. Production of ATP

C. Storage of glucose

D. Absorption of light

33. How does water travel up the xylem vessel on a hot, sunny day?

A. Mass flow under tension with cohesion between water molecules

B. Diffusion from higher to lower water concentration

C. Osmosis from roots to leaves

D. Active transport with energy provided by the Sun

34. In growing shoots, auxin is transported from the apex down the shoot. Proton pumps store energy in the form of a proton gradient and membrane potential, and then tap this energy source to allow the movement of auxin. What method of transport is this?

A. Diffusion

B. Active transport

C. Facilitated diffusion

D. Osmosis

35. Which chemicals are required for the germination of seeds to occur?

I. Water

II. Oxygen

III. Carbon dioxide

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

36. In a species with a diploid number of 16, what will the number of chromosomes be in a daughter cell after division by mitosis or meiosis?

	Mitosis	**Meiosis**
A.	16	16
B.	16	8
C.	8	16
D.	8	8

37. What causes genetic variation in gametes during meiosis?

A. Crossing over in prophase I and random orientation of homologous chromosomes in metaphase I

B. Crossing over in metaphase I and random orientation of homologous chromosomes in metaphase II

C. Linkage of genes in prophase I and crossing over in metaphase I

D. Linkage of genes in metaphase I and random orientation of homologous chromosomes in metaphase II

38. The Rh^+ antigen is found on the surface of red blood cells in people who are rhesus positive. A rhesus negative woman gives birth to a rhesus positive baby. What is a possible explanation for subsequent pregnancies triggering an immune response?

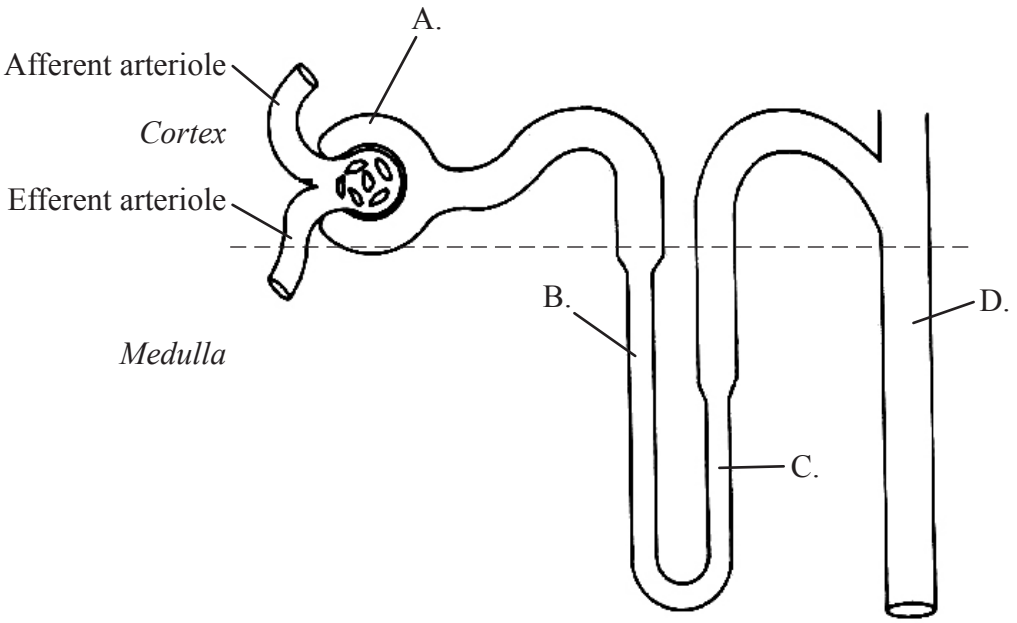
A. Exposure to the Rh^+ antigen in the first pregnancy triggered the development of specific phagocytes that could attack the blood of a future Rh^+ baby.

B. Exposure to the Rh^+ antigen in the first pregnancy triggered the development of antibodies that could attack the blood of a future Rh^+ baby.

C. The mother's immune system has been weakened by pregnancy.

D. Antibodies against the Rh^+ factor from the fetus pass to the mother across the placenta.

39. The diagram shows a nephron in a kidney. Which labelled part is permeable to sodium and not to water?



[Source: www.medcyclopaedia.com]

40. Which is the correct sequence of stages in fertilization?

A. cortical reaction → penetration of the egg membrane → acrosome reaction

B. cortical reaction → acrosome reaction → penetration of the egg membrane

C. acrosome reaction → cortical reaction → penetration of the egg membrane

D. acrosome reaction → penetration of the egg membrane → cortical reaction